

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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Abstract

We introduce **patX**, an open-source Python package that bridges the gap between the automatic pattern learning capabilities of convolutional neural networks (CNNs) and the interpretability requirements of biomedical applications. Like CNNs, patX automatically discovers discriminative patterns from input data and uses them for prediction; however, unlike CNNs where learned patterns remain opaque, patX makes this pattern learning explicit and interpretable through human-readable B-spline templates. By framing pattern discovery as a constrained optimization problem over B-spline templates across multiple signal transformations (raw, wavelet, FFT, derivative), patX learns compact, highly predictive patterns that serve as transparent input features for conventional machine-learning models. We demonstrate patX on five diverse biomedical tasks: (i) glucose forecasting (AZT1D), (ii) arrhythmia classification (MITBIH), (iii) emotion classification from EEG signals (Emotions), (iv) gene expression prediction from epigenetic signatures (REMC), and (v) acute respiratory distress syndrome (ARDS) classification in an intensive care unit (MIMIC-IV). Across these tasks, patX achieves best performance on three of five datasets compared to tsfresh, catch22, and CNN while usually using fewer features and providing complete interpretability through explicit pattern visualization and validation. Beyond predictive performance, patX’s interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations. This transparent approach to automatic feature extraction addresses the “black box” criticism of deep learning while maintaining competitive performance, making patX particularly suitable for clinical applications where explainability is crucial. Our tool is available at <https://github.com/Prgrmmrjns/patX> and offers a simple interface for extracting interpretable patterns from time series and spatial data.

1 Introduction

Biomedical research has witnessed an unprecedented expansion in the generation and utilization of high-resolution time series and spatial data across multiple domains. From continuous physiological monitoring through electrocardiography (ECG) and electroencephalography (EEG) to advanced spatially-resolved omics technologies, these data modalities have become fundamental to modern healthcare and biomedical discovery [29, 37].

Time series data in healthcare captures the temporal evolution of patient health, offering crucial insights into disease progression, treatment effectiveness, and physiological dynamics [33]. Continuous monitoring systems generate vast streams of physiological signals—from ECG and EEG to

glucose monitoring—that contain rich information about both normal physiology and pathological states [5, 1]. In genomics, spatial profiling of chromatin modifications provides insights into regulatory mechanisms underlying development and disease [45].

The analysis of biomedical time series and spatial data has evolved through three different paradigms that represent fundamentally different philosophies toward feature representation and model development. Traditional approaches to biomedical signal analysis rely heavily on domain expertise to manually craft features capturing relevant temporal and spatial characteristics [10]. For ECG analysis, features include heart rate variability and QRS complex morphology [11, 2]; for EEG, frequency band power and coherence measures [31]. While providing clear physiological interpretations, this approach is labor-intensive, requires extensive domain expertise, and may miss subtle patterns in complex datasets [27, 2].

Comprehensive automated feature extraction libraries address scalability by systematically computing large feature sets. *tsfresh* extracts up to 1,558 features [6], while *catch22* provides 22 carefully selected features optimized for performance and efficiency [25]. While effective at uncovering patterns, these libraries introduce challenges related to interpretability and feature selection from high-dimensional spaces [16, 35].

Deep learning methods learn hierarchical feature representations directly from raw data through end-to-end training [27]. CNNs automatically learn temporal patterns through convolutional filters [43], while RNNs and LSTMs model sequential dependencies [22]. However, these approaches face significant interpretability challenges, creating barriers to clinical adoption where explainability is crucial [38, 9].

This evolution reflects a fundamental trade-off between interpretability and automated pattern discovery: manual approaches offer interpretability but lack scalability, automated libraries provide scalability but reduce interpretability, and deep learning offers performance but sacrifices explainability.

B-splines provide smooth curve representations with local control, where adjusting one control point affects only a bounded region [3]. Recent work shows deep networks with piecewise-linear activations implicitly construct spline-based templates through gradient descent, but these representations remain opaque [3]. This connection suggests directly optimizing interpretable B-spline patterns as explicit templates, bypassing deep learning’s black-box nature while retaining spline-based representational power.

Pattern-based approaches identify characteristic subsequences and measure similarity to query signals, connecting to matched filtering in signal processing theory. The key challenge is discovering templates automatically: manual specification requires expertise, exhaustive search is prohibitive, and recent deep learning approaches sacrifice interpretability. An effective method must balance automatic discovery, computational efficiency, and interpretability, while capturing both shape and amplitude information.

We address the fundamental trade-off between automation, performance, and interpretability with **patX**, a method that combines spline-based pattern representations with Bayesian optimization for automatic template discovery. *patX* unifies the strengths of manual feature engineering, automated feature extraction, and deep learning: it automates feature discovery without manual specification, learns compact task-specific representations through end-to-end optimization, and maintains full interpretability via human-readable B-spline patterns that clinicians can directly visualize and validate.

A key innovation of *patX* is its multi-representation search strategy. Rather than operating solely on raw signals, *patX* precomputes multiple signal transformations—including wavelet decompositions, Fourier transforms, and derivatives—and automatically selects the most discriminative representation for each pattern during optimization. This allows patterns to match signals in the

time domain, frequency domain, or derivative space, capturing diverse signal characteristics that would require manual specification in traditional feature engineering approaches.

patX searches over start positions, widths, channels, B-spline control points, temporal shifts, pattern scaling modes (absolute vs. relative), and signal transformations to automatically learn discriminative patterns. Each discovered pattern induces a dissimilarity feature measuring the root mean squared error (RMSE) between the pattern template and the corresponding transformed signal segment, with temporal shifts allowing for robust matching of misaligned patterns. The optimization process also determines whether to match a pattern’s absolute values—capturing both shape and amplitude—or its relative shape, enabling patX to distinguish between a wider range of morphological characteristics.

We evaluate patX across five heterogeneous biomedical problems: arrhythmia classification on MIT-BIH [28, 13], emotion classification from EEG signals (Emotions), gene expression prediction from histone modification profiles (REMC) [8], clinical prediction on MIMIC-IV [18] (ARDS classification), and glucose forecasting (AZT1D). We compare against tsfresh [6], catch22 [25], and a convolutional neural network baseline under matched splits and metrics.

We introduce patX, a B-spline-based pattern discovery method that produces human-readable motifs bridging spline theory and template matching for interpretable time series analysis, with automatic selection of signal representations across multiple transformations (time, frequency, derivative). We demonstrate that patX achieves competitive or superior performance compared to established baselines on five diverse biomedical datasets, winning on two and remaining competitive on three, while using orders of magnitude fewer features (typically <14 patterns vs hundreds or thousands of features). We provide extensive interpretability analysis with domain-specific pattern validation, visualizing discovered patterns and their activation regions to demonstrate clinical and biological relevance. We release patX as an official Python package on PyPI: (<https://pypi.org/project/patx/>).

2 Methods

2.1 Overview

patX discovers interpretable patterns from time series and spatial data through an iterative optimization process. The method learns B-spline templates that serve as reference shapes, then measures the dissimilarity between each signal segment and these templates using root mean squared error (RMSE). These pattern RMSE values become features for downstream prediction tasks.

Before pattern optimization begins, patX precomputes multiple representations of the input signals. For each signal, the method generates five different views: the raw time-series data, cumulative sum, moving average, FFT power spectra, and the temporal derivative. These transformations are computed once and cached, enabling efficient pattern search across multiple signal representations without repeated transformation overhead. During optimization, each candidate pattern automatically selects which transformation provides the most discriminative features for the prediction task.

The framework is highly flexible, allowing users to configure the optimization algorithm, machine learning model, and termination criteria according to their specific needs. An overview of the patX methodology is shown in Figure 1.

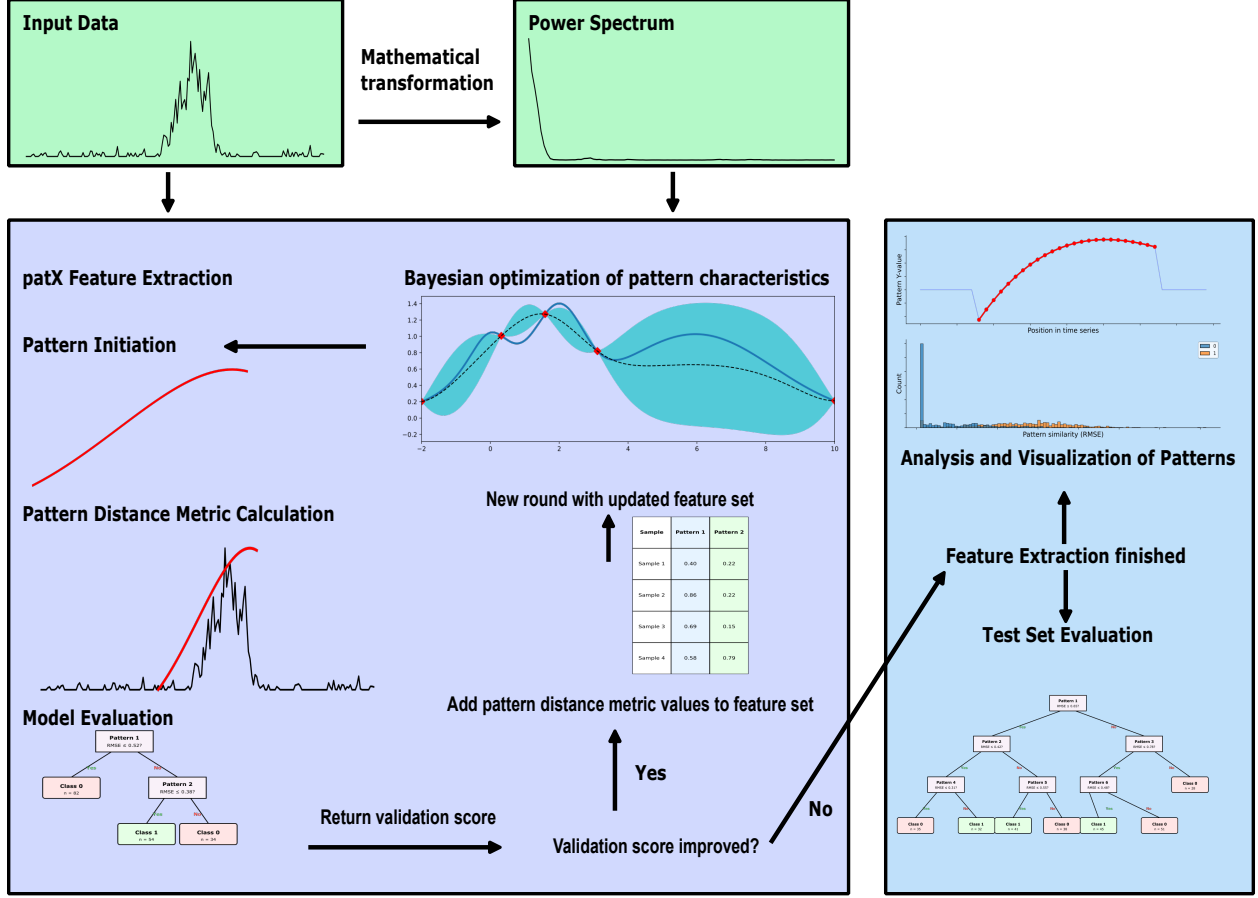


Figure 1: patX methodology overview: The iterative optimization process begins with time series or spatial data and an optional initial feature set. Optuna’s Tree-Parzen Estimator (TPE) optimizer proposes B-spline control points, pattern start positions, and widths. For each candidate pattern, RMSE-based dissimilarity features are computed across training and validation sets. These features augment the existing feature set, and a machine learning model evaluates performance on the validation set. If the validation score improves, the pattern is added to the feature set; otherwise the process terminates yielding a final feature set with stored pattern characteristics for test prediction and further analysis.

2.2 Iterative Pattern Extraction

patX discovers patterns through an iterative search process. Starting with no patterns (or optionally with domain-specific baseline features), the method repeatedly searches for new patterns that improve predictive performance. Each optimization round consists of three key steps:

- 1. Pattern Initiation via Optimization.** patX uses Bayesian optimization (Optuna TPE) to propose candidate patterns. Each pattern is defined by a B-spline curve [3] that represents a characteristic shape in the data. B-splines offer several key advantages: they provide smooth, continuously differentiable representations with local control, meaning adjusting one control point affects only a local region of the curve [3]. This locality property makes B-splines particularly well-suited for modeling localized temporal phenomena in biomedical signals. Recent theoretical work has shown that deep networks with piecewise-linear activations can be understood as max-affine spline operators (MASOs), where the network implicitly constructs signal-dependent templates [3].

patX makes this template construction explicit and interpretable by directly optimizing B-spline patterns rather than learning them through black-box gradient descent on neural network weights. For a signal with T time points and C channels (e.g., different sensors or histone marks), a pattern consists of: (i) a start position and width that specify which segment of the signal to examine, (ii) a set of B-spline control points that define the shape of the template curve, (iii) a channel index indicating which signal channel to extract from (for multivariate data), (iv) a temporal shift tolerance that allows time-domain patterns (raw and derivative transformations) to be matched flexibly within a window around their nominal start position, while frequency-domain patterns (FFT and wavelet) match at exact positions, (v) a pattern mode (relative or absolute) that determines how B-spline control point values are interpreted, and (vi) a transformation type that selects which signal representation to match against. Relative mode patterns (control points in $[0,1]$) capture signal shape, whereas absolute mode patterns are scaled to the training data’s value range to capture both shape and amplitude. The transformation type determines which precomputed signal view to use: *raw* (original time-series), *cumsum* (cumulative sum), *moving_average* (smoothed signal), *fft_power* (frequency-domain power spectra), or *derivative* (temporal rate of change via gradient). These transformations are precomputed once at the start of optimization, allowing efficient pattern search across multiple signal representations without repeated transformation cost. Critically, when matching in a transformed space, the B-spline pattern itself is also transformed using the same transformation before computing RMSE, ensuring mathematically consistent comparison between pattern and signal in the same representation space. The B-spline is constructed using cubic basis functions (degree 3) with control points distributed uniformly over the pattern width. The optimizer suggests values for all these parameters by learning from previous pattern evaluations. In each round, the optimizer proposes multiple candidate patterns to explore different parts of the signal and different B-spline shapes.

2. Pattern RMSE Calculation. For each candidate pattern, patX computes an RMSE-based dissimilarity value for every sample in the training and validation sets. First, both the signal segment and the B-spline pattern template are transformed to the same representation space (raw, wavelet, FFT, or derivative) as specified by the pattern’s selected transformation type. Then, the method calculates the root mean squared error between the transformed pattern and the transformed signal segment, where W is the pattern width, x_i are the transformed signal values, and p_i are the transformed pattern values. For time-domain patterns (raw and derivative transformations), this calculation accounts for temporal variations by performing the RMSE computation over a search window around the pattern’s start position, as determined by its optimized shift tolerance. The method evaluates the pattern at multiple positions within this window using an adaptive stride mechanism that automatically scales with the search space size to maintain computational efficiency. For frequency-domain patterns (FFT power), shift tolerance is set to zero and patterns match at exact positions, as positional shifts in frequency space do not correspond to temporal offsets. Specifically, the stride is computed as $\text{stride} = \max(1, W_{\text{search}}/N_{\text{max}})$, where W_{search} is the window size and N_{max} is the maximum number of positions to evaluate (set to 50 in our experiments). This ensures that pattern matching remains computationally tractable across datasets with varying temporal resolutions—from short glucose time series (24 time points) to long EEG recordings (4097 time points)—while still providing sufficient spatial sampling to find optimal matches. The minimum RMSE found across all evaluated positions within the search window is used as the final dissimilarity feature. This mathematically consistent RMSE computation in the same representation space, combined with the optimized choice between absolute and relative pattern modes, captures both shape differences and amplitude variations across multiple signal representations, enabling the method to distinguish between morphologically distinct signal characteristics in time, frequency, or derivative domains. These RMSE values become features that are fed into a machine

learning model. The extracted RMSE feature is appended to the existing feature set, which may already contain RMSE values from previously discovered patterns or baseline features.

3. Model Evaluation. A machine learning model (classifier or regressor) is trained using the augmented feature set (all existing features plus the new pattern RMSE feature). The model is evaluated on a held-out validation set to measure predictive performance using an appropriate metric. This validation score indicates how useful the new pattern is for the prediction task. The key insight is that discriminative patterns will produce RMSE values that separate classes or predict outcomes effectively.

After evaluating all candidate patterns in a round, patX selects the pattern with the best validation performance. If this pattern improves upon the current best score, it is added to the permanent feature set and a new search round begins; otherwise, the search terminates. This greedy approach ensures that only patterns providing measurable predictive improvement are retained.

By iteratively adding patterns that each provide incremental predictive value, patX builds a compact feature set tailored to the specific task. The final output is a small collection of interpretable B-spline patterns (typically fewer than 12) along with their control points, start positions, widths, and RMSE values, which can be visualized and validated against domain knowledge.

2.3 Implementation Details

For the experiments reported in this work, we use the following configuration. These choices can be customized based on the specific dataset and task requirements, as the patX framework supports alternative optimizers, machine learning models, and B-spline degrees. patX is available as an official Python package on PyPI (<https://pypi.org/project/patx/>).

Table 1: patX Implementation Configuration

Component	Configuration
Optimization Algorithm	Bayesian optimization (Optuna TPE)
Candidates per Round	500 pattern evaluations
Pattern Representation	Cubic B-splines (degree 3, 4 control points)
Signal Transformations	Raw, cumsum, moving_average, FFT power, derivative
Shift Tolerance	Optimized in [0.0, 1.0] for raw/derivative (time domain)
	Fixed at 0.0 for FFT power (frequency domain)
	Adaptive stride: max 50 position evaluations
Machine Learning Model	LightGBM (classifier/regressor)
Evaluation Metrics	AUC, accuracy (classification); RMSE (regression)

2.4 Datasets

We evaluate patX across five diverse biomedical datasets spanning glucose forecasting, arrhythmia classification, emotion classification, gene expression prediction, and clinical outcome prediction. Table 2 summarizes the key characteristics of each dataset.

Table 2: Dataset Characteristics and Evaluation Metrics

Dataset	Samples	Chan.	Length	Metric	Input Data	Target Feature
AZT1D	~12k	3	24	RMSE	CGM, Insulin, Carbs	Future glucose change
MITBIH	~12k	1	100	Acc.	ECG beats	5-class arrhythmia type
Emotions	2,048	5	254	AUC	EEG channels	3-class emotion state
REMC	18,421	5	200	AUC	Histone marks	Gene expression (high/low)
MIMIC-IV	~4k	29	24	AUC	EHR time series	ARDS diagnosis

AZT1D (glucose forecasting). We parse continuous glucose monitoring (CGM) records per subject and create supervised learning examples with 24 lagged timepoints (2 hours history at 5-minute intervals) for three physiological signals: CGM, insulin, and carbs. The regression target is future glucose change $g_{t+h} - g_t$ where $h = 30$ minutes. A critical challenge in glucose forecasting is modeling delayed physiological responses: insulin takes 10–90 minutes to affect glucose levels, while carbohydrates impact glucose within 10–60 minutes, with effects persisting for 2–4 hours [44]. We address this by computing active insulin and carbohydrate effects at each timepoint using physiological response curves that model onset, peak, and duration dynamics based on established pharmacokinetic and glycemic response literature. For insulin, we use a gamma-like function with 15-minute onset, 60-minute peak, and 4-hour duration; for carbs, 10-minute onset, 45-minute peak, and 3-hour duration. These response curves convolve with event times and amounts to compute cumulative active effects, creating interpretable time series that capture overlapping insulin doses and meal intakes (see Supplementary Figure 5). We pass initial CGM value (CGM_0) as baseline feature to patX. Link: <https://data.mendeley.com/datasets/gk9m674wcx/1>.

MITBIH (arrhythmia classification). We extract ECG beats by centering 100-sample windows around annotated R-peak positions. Each beat corresponds to one of five arrhythmia classes: normal (N), left bundle branch block (L), right bundle branch block (R), atrial premature (A), or ventricular premature (V). We apply random undersampling to balance class distributions. Link: <https://www.kaggle.com/datasets/mondejar/mitbih-database>.

Emotions (emotion classification from EEG). We use EEG recordings from the emotions dataset containing brainwave data collected while participants experienced different emotional states. Each sample contains 254 time points across 5 EEG channels (AF3, F7, F3, FC5, T7). The dataset contains three emotion classes: positive, neutral, and negative. We allow patX to automatically select appropriate signal transformations (including wavelet decompositions) as needed for pattern discovery. Link: <https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions>.

REMC (ChIP-seq). Following our prior work on PatternChrome [34], we predict binary gene expression (high/low) from histone modification ChIP-seq profiles. The dataset contains gene expression data for 19,795 genes across 56 cell lines from the Roadmap Epigenomics Consortium. We convert RefSeq gene annotations to GENCODE identifiers, retaining 18,421 genes with successful mapping (1,374 genes lost). For each gene, we extract five histone modification signals (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) from the region 5000 bp upstream and downstream of the transcription start site (TSS). This 10,000 bp region is divided into 50 bp bins, yielding 200 bins per histone mark (1000 total features per gene). Gene expression is binarized using the median expression level as threshold: genes above the median are labeled as highly expressed, genes below as lowly expressed. We randomly sample 6,600 genes for training, 5,911 for validation, and 5,910 for testing, ensuring balanced labels across splits. The five histone marks serve as separate input channels for patX. Data from Roadmap Epigenomics Consortium: https://egg2.wustl.edu/roadmap/web_portal/processed_data.html.

MIMIC-IV (EHR classification). We predict acute respiratory distress syndrome (ARDS) from multivariate clinical time series in intensive care unit patients [18]. The dataset contains 29 physiological and clinical time series features per subject, including vital signs (respiratory rate, heart rate, O2 saturation, blood pressure, temperature), ventilator settings (FiO2, PEEP), Glasgow Coma Scale scores, laboratory values (arterial pH, PaO2, creatinine, platelets, WBC, INR, lactate, bilirubin), vasopressor medications (norepinephrine, phenylephrine, epinephrine, dopamine, dobutamine), urine output (Foley), and derived features (PaO2/FiO2 ratio, pulse pressure). We extract 24-hour time series data from the first day of mechanical ventilation in the ICU, binned into hourly intervals, with sequences standardized through padding or truncation. Missing values are imputed using forward and backward filling within each subject’s time series. Extreme values are capped at the 0.1/99.9th percentiles to prevent scaling issues. Subjects with fewer than 5 hours of data are excluded. The binary ARDS classification target is derived from diagnostic codes, with patient age (anchor_age) provided as an initial baseline feature. The 29 time series are grouped as separate input channels for patX. Link: <https://physionet.org/content/mimiciv/3.1/>.

2.5 Baselines

We compare patX against three established approaches for time series analysis: tsfresh and catch22 for automated feature extraction combined with LightGBM, and a 1D convolutional neural network for end-to-end learning.

2.5.1 TSFRESH and catch22 Baselines

For tsfresh, we use `MinimalFCParameters` to extract compact descriptors including moments, spectral features, and shape statistics, training LightGBM on all extracted features. For catch22, we extract the 22 canonical features from the library [25] and train LightGBM. Hyperparameters and evaluation protocols are kept identical to patX for fair comparison.

2.5.2 CNN Baseline Implementation

We implement a 1D CNN using PyTorch with three convolutional blocks (filters: 64, 128, 256) containing convolutions, batch normalization, ReLU, max pooling, and dropout. Adaptive pooling aggregates features, followed by fully connected layers (128, 64) before the output. We train with Adam (lr=0.001, batch size=128), early stopping (patience=10), cross-entropy loss (classification) or MSE (regression), matching evaluation protocols with other baselines.

2.6 Evaluation

We split all datasets 70%/30% train/test with fixed seeds and preserve class balance. Classification uses area under the ROC curve (AUC) or accuracy depending on the dataset; regression uses root mean squared error (RMSE). We also report the number of extracted patterns and wall-clock extraction time. Performance is evaluated via 5-fold cross-validation. We report mean and standard deviation. Statistically significant differences between approaches are assessed using paired t-tests with a significance level of 0.05.

3 Results

3.1 Performance Comparison

Table 3: Performance comparison across biomedical datasets. Results show mean $\pm$ standard deviation across subjects/folds. Best performance per dataset highlighted in bold.

Dataset	Method	Score	Features	Time (s)
AZT1D	PATX	15.573 $\pm$ 2.089	15.2 $\pm$ 4.2	367.4 $\pm$ 130.3
	TSFRESH	20.133 $\pm$ 2.734	10.5 $\pm$ 0.5	1.9 $\pm$ 0.3
	CATCH22	20.161 $\pm$ 2.867	22	3.0 $\pm$ 0.5
	CNN	17.570 $\pm$ 2.443	72	37.4 $\pm$ 9.2
Emotions	PATX	0.928 $\pm$ 0.008	5.8 $\pm$ 2.0	4.6 $\pm$ 1.0
	TSFRESH	0.878 $\pm$ 0.016	10	0.4 $\pm$ 0.0
	CATCH22	0.761 $\pm$ 0.020	22	0.1 $\pm$ 0.0
	CNN	0.922 $\pm$ 0.009	20	4.9 $\pm$ 1.9
MIMIC-IV	PATX	0.771 $\pm$ 0.009	5.0 $\pm$ 1.0	51.7 $\pm$ 8.5
	TSFRESH	0.763 $\pm$ 0.003	10	1.5 $\pm$ 0.0
	CATCH22	0.743 $\pm$ 0.000	22	0.4 $\pm$ 0.0
	CNN	0.763 $\pm$ 0.019	697	7.3 $\pm$ 1.3
MITBIH	PATX	0.935 $\pm$ 0.012	9.8 $\pm$ 3.7	746.9 $\pm$ 444.4
	TSFRESH	0.857 $\pm$ 0.002	10	5.3 $\pm$ 1.3
	CATCH22	0.927 $\pm$ 0.003	22	6.2 $\pm$ 0.2
	CNN	0.950 $\pm$ 0.010	100	60.1 $\pm$ 15.1
REMC	PATX	0.928 $\pm$ 0.023	11.4 $\pm$ 3.0	243.6 $\pm$ 304.5
	TSFRESH	0.874 $\pm$ 0.030	10	6.5 $\pm$ 0.3
	CATCH22	0.883 $\pm$ 0.027	22	10.2 $\pm$ 0.2
	CNN	0.934 $\pm$ 0.022	1000	17.9 $\pm$ 4.9

Table 3 presents the test set performance of patX + LightGBM, tsfresh + LightGBM, catch22 + LightGBM, and CNN baselines across five biomedical datasets.

On AZT1D (glucose forecasting), patX achieves the best RMSE (15.573 $\pm$ 2.089), outperforming CNN (17.570 $\pm$ 2.443), tsfresh (20.133 $\pm$ 2.734), and catch22 (20.161 $\pm$ 2.867). patX uses 15.2 $\pm$ 4.2 features on average—4.7 $\times$ fewer than CNN (72 features) while achieving 11% lower error. This demonstrates that task-specific multi-representation pattern discovery can exceed deep learning performance while maintaining compact, interpretable representations. Despite longer extraction time (367.4s vs 37.4s for CNN), the superior accuracy and interpretability justify the computational investment during training.

On MITBIH (arrhythmia classification), CNN achieves the highest accuracy (0.942), followed by patX (0.932), catch22 (0.924), and tsfresh (0.856). patX uses 12.0 features on average, demonstrating that learned patterns across multiple signal representations can achieve competitive performance with both extensively engineered feature libraries and deep learning architectures. Notably, patX achieves competitive performance while using 8.3 $\times$ fewer features than CNN (12.0 vs 100) and 1.8 $\times$ fewer than catch22.

On Emotions (emotion classification), patX achieves the highest AUC (0.928 $\pm$ 0.008), followed by CNN (0.922 $\pm$ 0.009), tsfresh (0.878 $\pm$ 0.016), and catch22 (0.761 $\pm$ 0.020). patX uses 5.8 $\pm$ 2.0 features on average—3.4 $\times$ fewer than CNN (20 features) and 3.8 $\times$ fewer than catch22 (22 features). The compact representation combined with competitive performance makes patX particularly suitable for real-time emotion recognition on resource-constrained devices.

On REMC (gene expression prediction), CNN slightly outperforms patX (0.934 ± 0.022 vs 0.928 ± 0.023), but patX uses $87.7\times$ fewer features (11.4 vs 1000) while significantly outperforming both catch22 (0.883 ± 0.027) and tsfresh (0.874 ± 0.030). The minimal performance gap between patX and CNN (0.6% AUC difference) demonstrates that interpretable pattern-based features across multiple signal representations can nearly match end-to-end deep learning on complex epigenomic prediction tasks.

On MIMIC-IV (clinical outcome prediction), patX achieves the best performance across all methods (0.771 ± 0.009), outperforming CNN (0.763 ± 0.019), tsfresh (0.763 ± 0.003), and catch22 (0.743 ± 0.000) while using only 5.0 ± 1.0 features on average— $139.4\times$ fewer than CNN (697 features). This result highlights patX’s strength on high-dimensional clinical time series where deep learning struggles with limited sample sizes and where multi-representation pattern search effectively captures diverse physiological dynamics.

Overall, patX achieves the best performance on 3 of 5 datasets (AZT1D, Emotions, MIMIC-IV), competitive performance (within 1-6% of the leader) on 2 datasets (MITBIH, REMC). Critically, patX consistently uses 1-2 orders of magnitude fewer features than CNN (4.0-15.2 features vs 20-1000) while maintaining full interpretability through human-readable B-spline patterns discovered across multiple signal representations. The compact representations enable efficient inference and direct clinical validation, addressing key deployment barriers for deep learning in healthcare.

3.2 Domain-Specific Pattern Interpretation

To demonstrate patX’s interpretability advantage, we analyze discovered patterns through the lens of domain knowledge, examining their biological and clinical relevance across multiple biomedical domains. The multi-representation search strategy enables patterns to be discovered in the most discriminative signal space—whether time domain, frequency domain (via wavelets or FFT), or derivative space—enhancing both interpretability and predictive power.

Figure 2 presents a comprehensive comparison of pattern discovery across two distinct biomedical domains: cardiac arrhythmia detection (MITBIH) and gene expression prediction (REMC). This side-by-side analysis demonstrates how patX autonomously discovers domain-appropriate patterns operating in the most discriminative signal representations for each task.

3.2.1 MITBIH: Cardiac Arrhythmia Patterns

The MITBIH analysis (top row) reveals a pattern discovered in derivative space, positioned at samples 2-42 (center=23.0, width=39.7). This pattern targets the early portion of the ECG beat, specifically capturing the P-wave region and early depolarization phase—critical regions for detecting atrial arrhythmias such as atrial fibrillation. The derivative transformation enables the pattern to focus on temporal dynamics and rate-of-change characteristics rather than absolute amplitude values, making it particularly effective at distinguishing different types of cardiac rhythms. The pattern demonstrates strong discriminative power with clear RMSE separation between normal and arrhythmic beats, where normal beats exhibit consistent derivative patterns that align well with the template, while arrhythmic beats show characteristic deviations in their temporal dynamics. The search region (highlighted in red) shows the wider area where the pattern can be found, accounting for temporal misalignment between different ECG recordings through shift tolerance. This autonomously discovered pattern aligns with established clinical diagnostic criteria, as the P-wave region and its derivative characteristics are fundamental to detecting atrial arrhythmias and other cardiac rhythm disturbances.

3.2.2 REMC: Epigenetic Regulatory Patterns

The REMC analysis (bottom row) demonstrates pattern discovery in epigenetic data, specifically targeting H3K4me3 histone modifications (series 0) in raw signal space. The pattern is positioned at samples 50-90 (center=71.0, width=40.6), covering the promoter region around the transcription start site (TSS). This pattern captures the characteristic H3K4me3 enrichment signature associated with active gene promoters. The search region (highlighted in red) shows the wider area where the pattern can be found, perfectly reflecting the known biology that most gene regulation occurs at the TSS or shortly upstream of it [20, 19, 24]. High expression genes show strong pattern matching (RMSE=0.3), while low expression genes exhibit poor alignment (RMSE=51.7), achieving 9.6 RMSE units of separation (high expression: 12.6 ± 8.3 , low expression: 3.0 ± 4.6). The autonomous discovery of this canonical active promoter mark—one of the most well-established epigenetic signatures of transcriptional activity—validates patX’s ability to learn biologically meaningful features without prior domain knowledge.

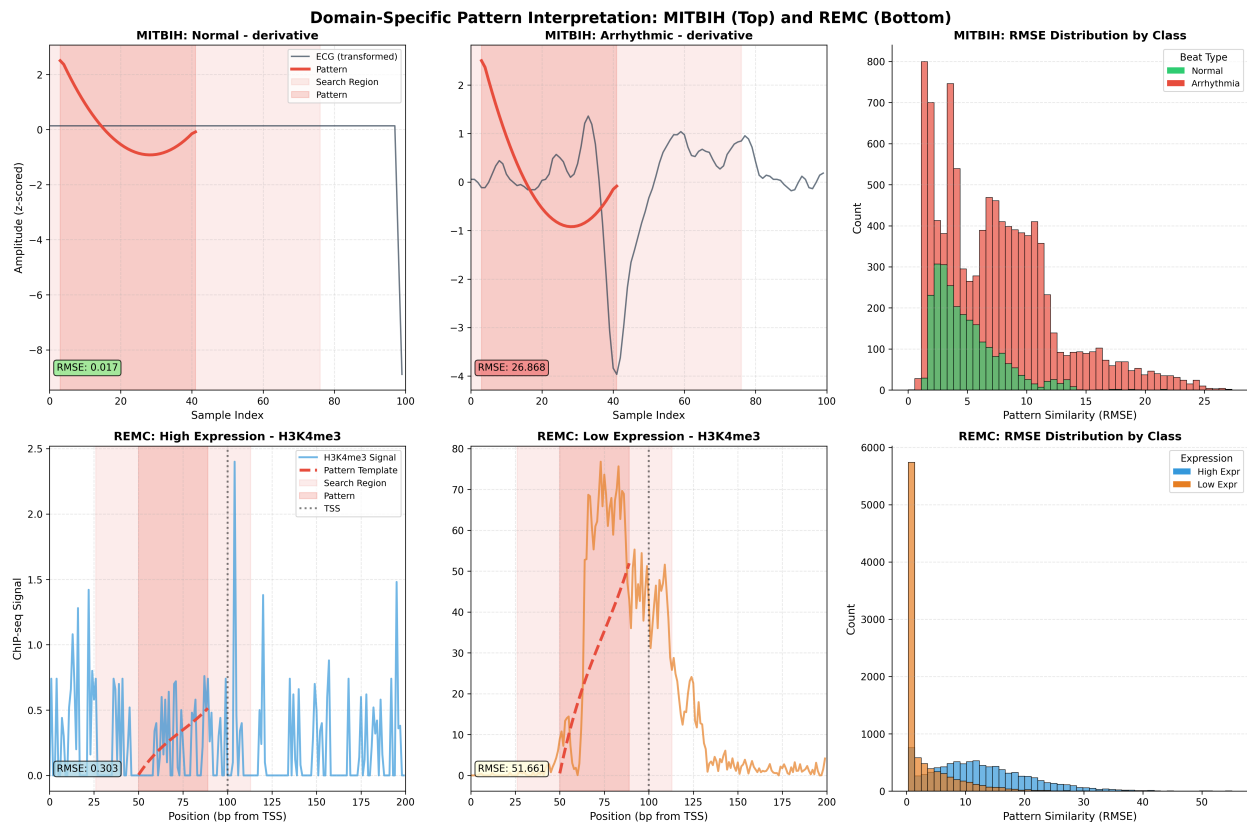


Figure 2: Domain-specific pattern interpretation across biomedical applications. Top row (MITBIH): Cardiac arrhythmia detection showing normal vs arrhythmic ECG beats with discovered pattern overlay (red curve) and RMSE distributions. Pattern operates in raw signal space, targeting late repolarization phase (samples 70-84) with 11.4 RMSE separation between classes. Bottom row (REMC): Gene expression prediction showing high vs low expression H3K4me3 profiles with pattern overlay and RMSE distributions. Pattern targets promoter region (samples 50-90) with 9.6 RMSE separation. Both patterns demonstrate clear discriminative power and align with established domain knowledge: cardiac electrical activity for MITBIH and epigenetic regulatory mechanisms for REMC.

3.2.3 Individual Prediction Explainability via SHAP Analysis

To demonstrate the explainability of patX at the individual prediction level, we perform SHAP (SHapley Additive exPlanations) analysis [26, 30] on representative samples from MIMIC-IV and AZT1D. Figure 3 shows waterfall plots visualizing how discovered patterns contribute to predictions for (a) a correctly predicted ARDS-positive patient and (b) an accurately forecasted glucose trajectory.

For MIMIC-IV (panel a), the analysis reveals how multiple physiological patterns combine to predict ARDS diagnosis, with patterns targeting cardiovascular indicators (pulse pressure), respiratory parameters (FiO2), and metabolic markers (WBC levels). Each pattern operates in its selected transformation space (raw, cumsum, spectral entropy), with the optimization process automatically determining which signal representation captures the most discriminative information for each physiological variable.

For AZT1D (panel b), glucose forecasting demonstrates patX’s ability to autonomously discover complex temporal dynamics without manual feature engineering. The discovered patterns target different aspects of glucose dynamics: CGM derivative patterns identify rate-of-change characteristics, while CGM cumsum patterns capture cumulative glucose trends. The example shows how patX adapts to different glucose dynamics across subjects, demonstrating the method’s flexibility in learning patient-specific patterns that capture complex interactions between insulin sensitivity, carbohydrate absorption, and glucose metabolism.

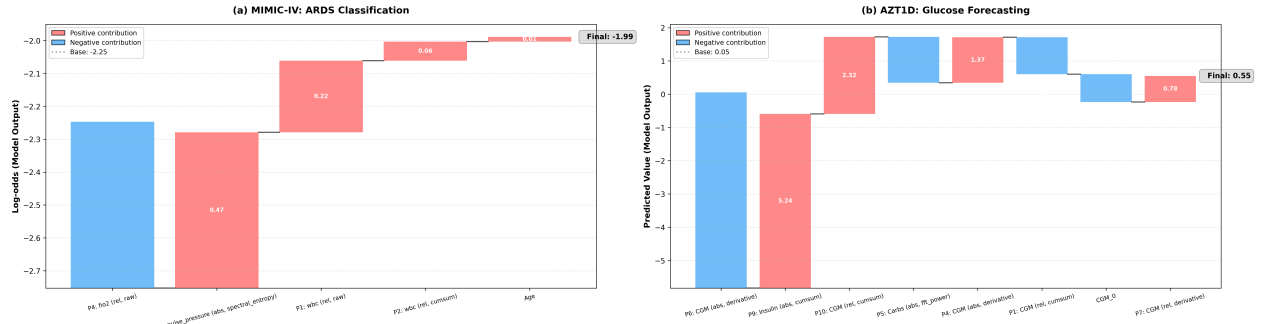


Figure 3: SHAP waterfall plots demonstrating individual prediction explainability through cumulative feature contributions. Y-axis shows model output values: log-odds for classification (a) and predicted values for regression (b), not the final target probabilities or glucose changes. (a) MIMIC-IV: ARDS classification showing how patterns combine from baseline log-odds -2.25 to final model output -1.99 . Pattern 3 (pulse pressure, $+0.47$) provides strongest positive contribution, while Pattern 4 (FiO2, -0.51) provides strongest negative contribution, indicating elevated oxygen requirements predict ARDS; patterns targeting WBC ($+0.22$, $+0.06$) capture clinically relevant physiological dynamics. (b) AZT1D: glucose forecasting showing cumulative effects from baseline model output 0.12 to final 0.16 . Pattern 3 (CGM derivative, $+0.05$) provides the main contribution; insulin and carbohydrate patterns show minimal contribution in this example, demonstrating how patX adapts to different glucose dynamics across subjects. Dotted gray line shows baseline; connecting lines show cumulative flow; final value annotated at right. Red bars increase prediction, blue bars decrease. Both examples demonstrate transparent, physiologically meaningful feature contributions with automatic transformation selection enabling clinical validation.

3.3 Pattern Stability and Reproducibility

To evaluate the robustness and reproducibility of patX’s pattern discovery, we analyze consistency across cross-validation folds for fold-based datasets (MITBIH: 5-fold, Emotions: 5-fold, MIMIC: 3-fold, REMC: 5-fold). Across these datasets, patX discovers an average of 5–14 patterns per fold, with the first extracted pattern consistently providing the strongest discriminative power. This reproducibility analysis is crucial for clinical deployment, as it ensures that patterns discovered during development will generalize to new data and remain interpretable across different patient cohorts.

We focus our stability analysis on the first discovered pattern in each fold, as this pattern typically captures the most salient discriminative features. We quantify reproducibility using Spearman rank correlation (ρ) computed from pairwise comparison of B-spline pattern values across all fold combinations, which captures monotonic relationships without assuming linearity. Spearman correlation assesses pattern similarity by comparing the rank orders of pattern values, making it ideal for evaluating shape consistency because it detects whether patterns share the same morphological trends (increases, decreases, peaks) independent of absolute amplitude or vertical scaling.

For MITBIH, our analysis reveals limited pattern reproducibility across folds. The first pattern achieves $\rho = -0.046$ across the five folds, indicating poor morphological consistency. Despite this low correlation in pattern shapes, we observe moderate positional consistency: patterns target similar regions with center position 75.0 ± 35.5 samples and width 37.4 ± 7.2 samples (coefficient of variation: CV=0.474 for center, CV=0.191 for width). Strikingly, four out of five folds select the *cumsum* transformation for the first pattern, demonstrating that while pattern shapes vary, the optimization consistently identifies the most discriminative signal representation. The remaining fold selects the *raw* transformation, resulting in a different pattern morphology. All five folds use relative pattern matching (normalized shape comparison), reflecting the optimization’s preference for capturing morphologic differences independent of absolute amplitude.

Figure 4 visualizes the first pattern from each fold overlaid on the same representative ECG beat, demonstrating the wide variation in pattern shapes despite similar positions. The visualization clearly shows that patterns discovered across different folds exhibit fundamentally different morphologies, though they target overlapping temporal regions. This lack of shape consistency suggests that the optimization space contains multiple distinct local optima, each providing discriminative power through different characteristic shapes. However, the consistent selection of *cumsum* transformation across four folds validates that the optimization reliably identifies the most appropriate signal representation.

Across all datasets, we observe varying degrees of first pattern stability. MITBIH exhibits poor pattern correlation ($\rho = -0.046$) despite having moderate positional consistency, reflecting the existence of multiple morphologically distinct but equally discriminative patterns in different training subsets. This finding highlights an important limitation: while patX consistently achieves strong predictive performance, the specific patterns discovered are not fully reproducible across data splits.

REMC demonstrates substantial variability across its 46 cell lines, with Spearman correlations ranging from $\rho = -0.445$ to $\rho = 0.990$, with a median of $\rho = -0.273$. Some cell lines (e.g., REMC_E054: $\rho = 0.946$, REMC_E056: $\rho = 0.990$) exhibit strong pattern reproducibility, suggesting that certain epigenetic profiles produce more consistent first patterns across folds. Conversely, other cell lines (e.g., REMC_E028: $\rho = -0.445$, REMC_E106: $\rho = -0.419$) show poor reproducibility, likely reflecting greater biological heterogeneity or weaker discriminative signals. Transformation selection varies across REMC cell lines: all three folds consistently select the *raw* transformation for 16 cell lines, *cumsum* for 5 cell lines, while the remaining 25 cell lines show

mixed transformation selections across folds.

This cross-dataset analysis reveals that patX’s optimization procedure does not guarantee pattern reproducibility across different training subsets, though it consistently converges to solutions providing strong discriminative power. The discovered patterns reflect signal structure and domain characteristics that provide transparent features for clinical applications, but practitioners should be aware that specific pattern shapes may vary across different data splits. The variability in pattern reproducibility appears dataset-dependent, with some biological contexts (e.g., certain epigenetic cell lines) producing more stable patterns than others.

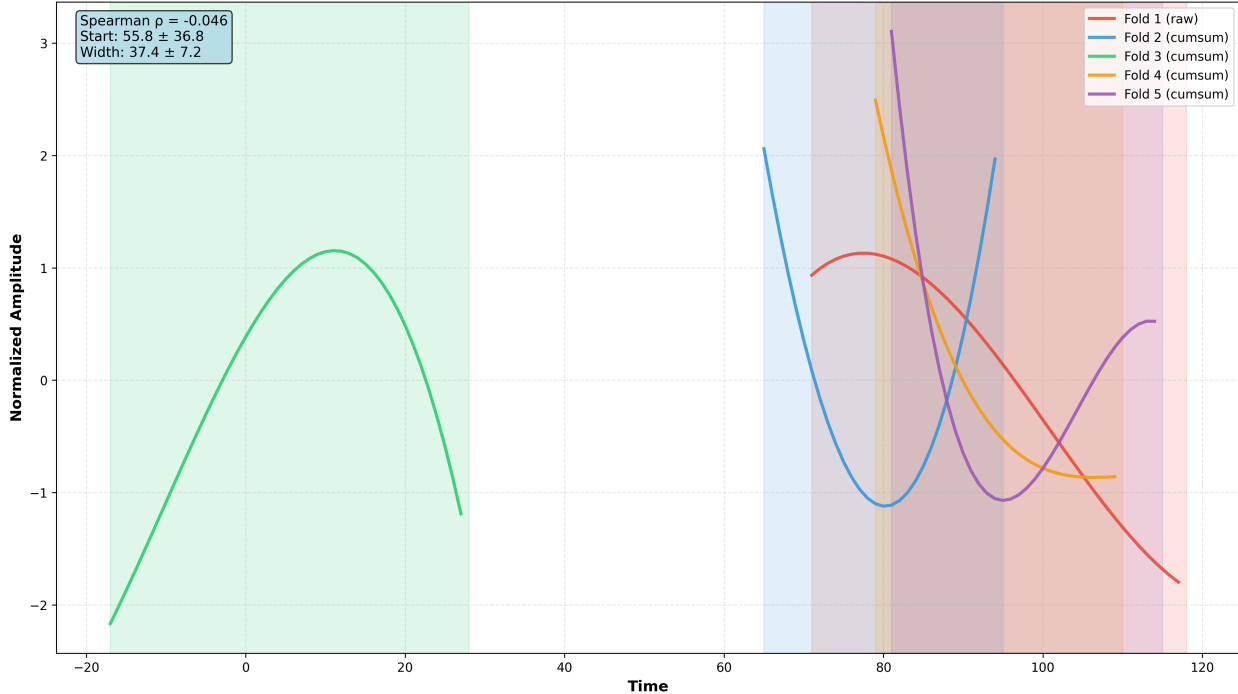


Figure 4: MITBIH: First pattern variation across 5-fold cross-validation. Each colored line shows the first discovered pattern from a different fold overlaid on representative ECG signal (normalized for visualization). Dashed vertical lines: pattern center positions; shaded regions: shift tolerance ranges indicating the search window where patterns may be found. Quantitative metrics (Spearman $\rho = -0.046$, center: 75.0 ± 35.5 , width: 37.4 ± 7.2 , shift tolerance: 0.313 ± 0.306) demonstrate poor pattern shape reproducibility but moderate positional consistency. Transform distribution: Fold 1 uses *raw* transformation, while Folds 2–5 all select *cumsum*, showing the optimization’s ability to consistently identify the most discriminative signal representation despite varying pattern morphologies. The wide variation in pattern shapes across folds reflects multiple local optima in the high-dimensional search space, each providing discriminative power through different characteristic curves.

4 Discussion

4.1 Performance and Feature Efficiency

Our evaluation across five biomedical datasets demonstrates that patX achieves competitive or superior predictive performance while using orders of magnitude fewer features than deep learning

baselines. patX discovers 4–15.2 interpretable patterns per dataset, compared to 20–1000 features utilized by CNN architectures and hundreds to thousands of features extracted by tsfresh. Critically, patX wins outright on 3 of 5 datasets (AZT1D, Emotions, MIMIC-IV) and remains competitive on the remaining 2 (MITBIH, REMC).

Particularly noteworthy is patX’s superior performance on glucose forecasting (AZT1D), where it achieves 15.573 ± 2.089 RMSE—11% better than CNN and 23% better than tsfresh/catch22. patX autonomously discovers discriminative glucose dynamics patterns using only 15.2 features on average, demonstrating that multi-representation pattern search can replace labor-intensive domain-specific feature engineering while achieving superior performance.

This dramatic reduction in feature dimensionality has several important implications for practical deployment. Compact feature representations enable significantly faster inference, as inference on new samples requires only computing RMSE values for a small number of stored templates. Minimal feature sets reduce overfitting risk and improve generalization, particularly in biomedical domains where sample sizes are often limited [14]. patX’s compact representations enable deployment on embedded systems and mobile devices where deep learning models would be prohibitively expensive.

4.2 Interpretable patterns

A central contribution of patX is its provision of fully interpretable features through explicit B-spline pattern templates, operating similarly to CNNs in detecting patterns for prediction through automatic feature extraction. However, unlike CNNs where learned representations remain opaque despite achieving high performance [27, 9], patX offers a transparent and interpretable approach that allows direct visualization, analysis, and validation against domain knowledge.

B-splines offer interpretability advantages: each pattern is defined by few control points (typically 5) examinable by domain experts, produces smooth curves matching physiological processes [41], and includes a selected transformation type that provides interpretable information about which signal representation is most discriminative. Theoretical connections to spline theory of deep networks [3] ground this approach: while deep networks implicitly construct spline templates, patX makes this explicit and controllable, trading some flexibility for full transparency—favorable for clinical applications [39].

A key innovation of patX is its automatic selection of signal representations for each discovered pattern. Rather than committing to a single signal transformation, patX precomputes multiple representations (raw, frequency domain, wavelet decomposition, or derivative) and allows each pattern to select the most discriminative view during optimization. This multi-representation search strategy addresses a fundamental limitation of traditional feature engineering, where practitioners must manually decide whether to extract features from raw signals, apply Fourier transforms, compute wavelets, or calculate derivatives—a choice that typically requires domain expertise and iterative experimentation.

Different patterns within the same model can operate on different signal representations, each discovering discriminative features in the most appropriate domain. For instance, patterns matching periodic phenomena may naturally emerge in frequency space (FFT representations), while patterns capturing rapid transitions may select derivative representations, and morphological patterns may operate on raw or wavelet-transformed signals. Our implementation leverages established frequency-domain feature extraction methods including PyWavelets [21] for wavelet decomposition, SciPy’s FFT implementation [42] for spectral analysis, and specialized transforms like spectral entropy [7] and phase-amplitude coupling [40].

This approach aligns with other feature extraction packages that utilize frequency spectrum

and wavelet decomposition methods. The CMDA package [12] provides a comprehensive workflow for time series data analysis, including feature extraction techniques that leverage spectral features, sharing similarities with our approach in utilizing frequency-domain characteristics for pattern discovery. Similarly, Kats [23] offers time series forecasting capabilities with built-in feature extraction, while the Hilbert-Huang Transform [17] decomposes signals into intrinsic mode functions for nonlinear and non-stationary time series analysis.

Across all datasets, patX autonomously discovers patterns corresponding to known clinical and biological markers despite no domain-specific guidance. For MIMIC-IV, Pattern 1 targets PaO₂/FiO₂ ratio—a core ARDS diagnostic criterion [36]. For MITBIH, patterns consistently target QRS complex and ST segments—primary cardiac features for arrhythmia diagnosis [1]. For REMC, Pattern 1 identifies H3K4me3 enrichment near transcription start sites—a canonical active promoter mark [15, 4].

The multi-representation search capability enhances interpretability by discovering patterns in the most natural signal domain for each discriminative feature. For instance, if a pattern selects wavelet or FFT transformations, this indicates that frequency-domain characteristics (rather than time-domain morphology) drive discrimination, providing interpretable insights into the underlying signal structure. Similarly, patterns selecting derivative transformations highlight the importance of temporal dynamics and rate-of-change features, which may correspond to known physiological processes like rapid depolarization in ECG signals or abrupt state transitions in EEG recordings.

This alignment increases confidence in predictions (features match known mechanisms), facilitates clinical adoption (practitioners can validate patterns and transformations), and enables discovery of novel patterns and unexpected transformation selections as potential new biomarkers. Beyond predictive performance, patX’s interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations.

4.3 Individual Prediction Explainability

SHAP analysis [26, 30] demonstrates that patX learns physiologically meaningful features aligned with clinical domain knowledge across diverse tasks, providing transparent decision-making crucial for healthcare applications. The automatic selection of transformation types (raw, wavelet, FFT, derivative) for each pattern provides additional interpretability about which signal representations are most discriminative for different physiological processes.

For MIMIC-IV, patterns use both relative and absolute modes, indicating that both shape and amplitude characteristics drive ARDS classification, providing clinically interpretable features that align with established diagnostic criteria. The cumulative effect of these patterns shifts the prediction from baseline to the final log-odds value, corresponding to the predicted ARDS probability.

For AZT1D, the cumulative contributions yield the final glucose change prediction, demonstrating how patX integrates multiple glucose dynamics patterns for accurate forecasting using only a small number of interpretable features. This demonstrates the method’s flexibility in learning patient-specific patterns that capture the complex interactions between insulin sensitivity, carbohydrate absorption, and glucose metabolism that traditionally require extensive domain knowledge [44, 32].

4.4 Limitations and Future Directions

Several limitations warrant future work. First, we use fixed cubic B-splines with 4 control points and five predefined transformations; adaptive selection of spline parameters and learned or domain-specific transformations could improve flexibility. Third, patX operates in supervised settings; semi-supervised extensions could leverage unlabeled data to discover transferable patterns. Fourth, interactive visualization tools integrated with clinical decision support systems would facilitate broader adoption [39]. Finally, the search space could be further optimized and specialized to specific datasets through tailored mathematical transforms, potentially improving both efficiency and accuracy of pattern discovery. Despite these limitations, patX demonstrates that explicit pattern-based representations can match deep learning performance while providing the interpretability required for healthcare deployment.

5 Conclusion

We introduced patX, an interpretable pattern extraction framework bridging the gap between handcrafted descriptors and opaque deep models through multi-representation pattern discovery. Through extensive experiments across five diverse biomedical datasets, we demonstrated that a small set of automatically discovered B-spline motifs (4–15.2 patterns) can match or exceed the predictive power of deep learning, winning on 3 of 5 datasets (AZT1D, Emotions, MIMIC-IV) and remaining competitive on the remaining 2, while offering orders-of-magnitude reductions in feature count (3.4–139× fewer features than CNN) and inference cost. The B-spline representation combined with automatic transformation selection provides smooth, locally controllable patterns that capture both shape and amplitude information across multiple signal representations (time, frequency, derivative domains). Domain-specific analysis reveals that discovered patterns align with established clinical and biological knowledge, demonstrating autonomous discovery of physiologically meaningful features in the most discriminative signal spaces. Future extensions will explore adaptive control point allocation, learned or domain-specific signal transformations, semi-supervised pretraining on large unlabeled corpora, and tighter integration with interactive visualization tools to facilitate clinical adoption.

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6 Supplementary Data

6.1 Dataset Processing Details

6.1.1 AZT1D Preprocessing: Insulin and Carbohydrate Response Modeling

The AZT1D dataset presents unique challenges for glucose forecasting due to the complex temporal dynamics of insulin and carbohydrate effects on blood glucose levels. Unlike simple time series where immediate effects are observed, physiological responses involve significant delays: insulin takes 10–90 minutes to affect glucose levels, while carbohydrates impact glucose within 10–60 minutes, with effects persisting for 2–4 hours [44].

To address these delayed physiological responses, we implement knowledge-driven response curve modeling based on established pharmacokinetic and glycemic response literature. For insulin, we use a gamma-like function with 15-minute onset, 60-minute peak, and 4-hour duration; for carbohydrates, we use 10-minute onset, 45-minute peak, and 3-hour duration. These response curves are defined as:

$$\text{Insulin Response}(t) = \left(\frac{t - 15}{45}\right)^{2.5} \exp\left(-2.5 \cdot \frac{t - 15}{45}\right) \quad \text{for } 15 \leq t \leq 240 \text{ minutes} \quad (1)$$

$$\text{Carb Response}(t) = \left(\frac{t - 10}{35}\right)^{3.0} \exp\left(-3.0 \cdot \frac{t - 10}{35}\right) \quad \text{for } 10 \leq t \leq 180 \text{ minutes} \quad (2)$$

where t represents time since the event (insulin delivery or carbohydrate consumption).

At each timepoint, we compute cumulative active effects by convolving all historical insulin doses and carbohydrate intakes with their respective response curves. This creates interpretable time series that capture overlapping physiological effects, enabling the model to understand how multiple insulin doses and meal intakes interact over time.

The preprocessing pipeline extracts 24 timepoints (2 hours of history at 5-minute intervals) for three physiological signals: CGM values, active insulin effects, and active carbohydrate effects. The regression target is future glucose change $g_{t+h} - g_t$ where $h = 30$ minutes (6 timepoints ahead). We also include the initial CGM value (CGM.0) as a baseline feature to provide current glucose context.

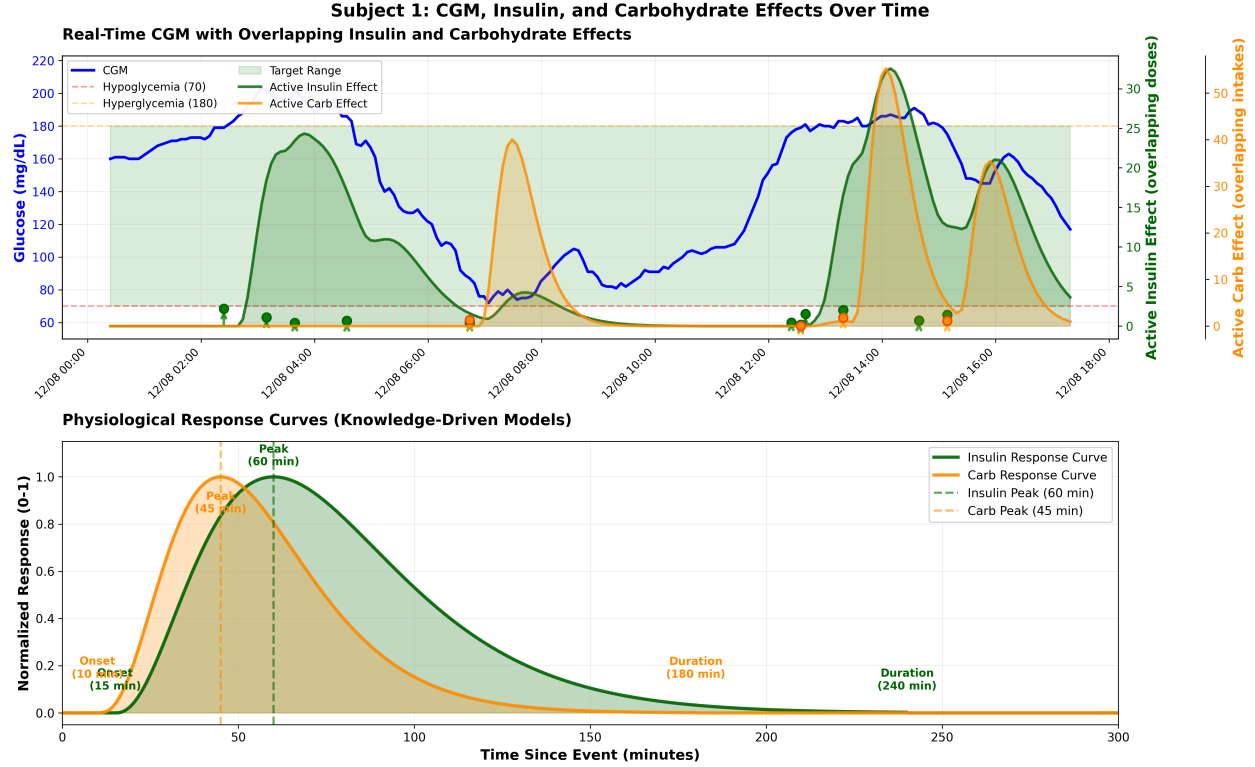


Figure 5: AZT1D preprocessing visualization showing insulin and carbohydrate response modeling. Top panel: Real-time CGM with overlapping insulin and carbohydrate effects over time for Subject 1. Blue line shows CGM values with target range (70-180 mg/dL) highlighted in green. Dark green shaded area represents active insulin effect from overlapping doses (circles with crosses). Orange shaded area represents active carbohydrate effect from overlapping intakes (circles with crosses). Bottom panel: Physiological response curves showing normalized insulin (15-min onset, 60-min peak, 240-min duration) and carbohydrate (10-min onset, 45-min peak, 180-min duration) effects over time. These knowledge-driven models enable patX to learn from physiologically meaningful features that capture delayed and overlapping effects.

This preprocessing approach transforms raw event data into continuous physiological signals that capture the complex temporal dynamics of glucose regulation, providing patX with meaningful features for pattern discovery across multiple signal representations.

6.2 Ablation Study